Accurate Promoter Models Only Partly Learn TATA-Box Position: An Audit With Architecture-Matched Null Models

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Code and data: https://github.com/Maheshbonthada/promoter-grammar-gcr

Accurate Promoter Models Only Partly Learn TATA-Box Position: An Audit With Architecture-Matched Null Models

Sravan Kumar Bonthada

***Abstract*—DNA sequence models are used to prioritise non-coding variants, and their accuracy is often read as evidence that they have learned regulatory mechanism. We test this for the TATA box, whose position near -30 is set by the structure of its protein–DNA complex. Positional use is measured by grammar discrimination (GD), built from block swaps that preserve base composition, so that any mononucleotide-composition scorer obtains exactly 0.5. Because GD has no universal chance level, each architecture is retrained with every canonical-TATA promoter removed; these null models score 0.48 to 0.80. Against their own nulls, six models with test AUROC of 0.940 to 0.962, including a 495M-parameter genomic language model, capture 2% to 69% of the room above their null, and a published architecture, DeepSTARR, captures 52%. The shortfall is not chiefly due to base composition: the TATA motif is anti-predictive on this benchmark (AUROC 0.212), but with composition-matched negatives the baselines still captured only 13% to 30%. A ranking constraint over the same counterfactuals shows that the gap can be closed, capturing 90% to 99% in the six architectures for a change in test AUROC of at most 0.006, but it failed for DeepSTARR. Against measured variant effects, correct positional rules outperform displaced ones in four architectures but, after multiple-testing correction, improve on the unconstrained model in only one. On an independent lentiMPRA dataset the audit replicates, and the constraint costs 3% of predictive correlation at a reduced weight and 11% at the default weight.**

***Index Terms*—Regulatory genomics, deep learning, model interpretability, shortcut learning, counterfactual explanation, transcription factor binding, null models.**

I. Introduction

SEQUENCE-to-function models are now used to interpret non-coding variation, including variants in promoters whose disruption causes disease. The inference such use depends on is mechanistic: that a model which predicts regulatory labels accurately has learned something about how regulatory sequence works, and that its response to a mutation therefore carries information about that mutation's consequence. That inference is rarely tested.

It is not safe by default. Machine-learning systems reach high accuracy through whatever feature is most predictive in the training distribution, including features that are artefacts of how the data were assembled [1]. In regulatory genomics the obvious candidate is base composition. Human promoters are GC-rich relative to bulk genomic sequence, so a classifier separating promoters from genomic background can succeed on composition alone, without representing any binding site, spacing constraint or positional rule.

The gap matters because position is not incidental to regulatory function. The TATA box is recognised by the TATA-binding protein (TBP), which bends DNA sharply and positions the pre-initiation complex; the resulting constraint places the element approximately 30 bp upstream of the transcription start site (TSS) [2], [3], [4], a position evident both in early weight-matrix analyses of promoter elements [5] and in curated promoter collections [6]. Mutations that disrupt this element cause disease: β-thalassemia arises from single substitutions in the *HBB* TATA box [7]. A model whose predictions are used to interpret such variants, but which does not use the element's position, is not interpreting the mechanism it appears to.

The TATA box is not a universal feature of promoters. Only a minority of human core promoters carry one [8], and in the benchmark used here 392 of 13,611 training promoters (2.9%) carry a canonical TATA box at -30. A promoter-versus-background classifier therefore has little reason to learn it, and its failure to do so would be unremarkable if such models were used only to classify promoters. It matters because the same models are used to score variants in promoters that do depend on the element, of which the *HBB* TATA box is the clearest case. The question this paper asks is whether their scores for such variants reflect the element at all.

Existing evidence is mixed and indirect. Pretrained genomic language models add little over one-hot encodings on many regulatory tasks [9]; sequence-to-expression models capture some regulatory determinants and miss others [10]; in-silico perturbation can reveal what a model has learned [11], [12] and is routinely used to interpret large sequence-to-expression models [13], although gradient-based attributions themselves need correction before they can be read reliably [14]; and base-resolution binding models can learn motif spacing [15]. Models trained to predict transcription initiation at base-pair resolution learn motifs with position-specific effects [16], including the roughly 30-bp spacing of the TATA box upstream of the TSS [17], so whether a model learns this grammar depends on what it is trained to predict. The question here concerns the classification and activity models that are also used to score variants. Benchmarks for DNA language models have also been criticised for tasks that do not reflect biological difficulty [18]. What has been missing is a direct, composition-controlled test of whether accurate promoter models use positional grammar, together with a reference for what score

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All experiments ran on a single NVIDIA RTX 3050 with 8 GB of memory.

a model with no such knowledge would obtain.

Training a model to respect a known rule is not new. Counterfactually augmented data teaches a classifier which change in an input should change its label [19]; "right for the right reasons" and attribution priors penalise a model for placing gradient mass on features an expert marks irrelevant [20], [21]; EvoAug applies evolution-inspired augmentations to genomic networks [22]; and composition-matched backgrounds, such as the dinucleotide-shuffled negatives used in motif learning [23] or dedicated background generators [24], address the same shortcut in the dataset rather than the loss. The constraint used here differs in three respects that matter for this paper. Its counterfactuals preserve the window's base composition exactly, so the constraint cannot be satisfied by composition and the measure that evaluates it is composition-blind by construction. It constrains the ordering of scores between a sequence and its counterfactual rather than an attribution map, so it requires no gradient-attribution method and asserts nothing about the counterfactual's label. And the rule it enforces is positional: a named element must matter at a named offset. Positional effects of this kind have been reported in plant regulatory sequence [25]; we are not aware of a prior composition-controlled test of whether human promoter classifiers use them, nor of a chance level against which such a test could be read.

This paper supplies both. Its main contribution is the audit and the null-model methodology the audit requires; a training constraint is then evaluated as a demonstration that the gap can be closed, not as a general remedy.

First, a composition-controlled measure. Grammar discrimination compares the effect of displacing the TATA box with the effect of displacing neighbouring blocks in the same promoter, using block swaps that preserve the window's base composition exactly. Any scorer that depends on the window only through mononucleotide composition therefore obtains GD of exactly 0.5. This follows directly from the construction and is not presented as a result; its use is that $\mathrm{GD} > 0.5$ excludes a mononucleotide-composition explanation. Because block swaps change the dinucleotides at the block junctions, GD is not blind to higher-order k-mer composition, and we report how far k-mer scorers depart from 0.5.

Second, a reference. We show that GD has no universal chance level: a model trained with every canonical-TATA promoter removed can still score 0.80. That is possible because the TATA block differs from its neighbours in local composition, so a model can separate the two edits without representing the motif. Positional scores are therefore uninterpretable unless each model is compared with its own architecture-matched null.

Third, the audit. Against their own nulls, six accurate models capture between 2% and 69% of the room above the null. Scoring the TATA motif on this benchmark classifies promoters at AUROC 0.212: the element is anti-predictive, because the negatives are AT-rich relative to GC-rich promoters. That penalty is not the main explanation. Retraining with composition-matched negatives, which removes it, did not close the gap, and raising the element's frequency in training closed it only in part. A published architecture, DeepSTARR, retrained on the same data, behaves like the others.

Fourth, a demonstration that the gap can be closed. Grammar-Consistency Regularization (GCR) adds a ranking constraint requiring each native sequence to score above composition-preserving counterfactuals in which the protected element is displaced. We evaluate it against falsification controls in which the same loss enforces deliberately wrong rules, against external measurements never used in training, and on an independent dataset of measured regulatory activity, and we report where its benefit stops, including an architecture for which it fails.

II. Materials and Methods

A. Data

Promoter classification used the revised Nucleotide Transformer benchmark (`InstaDeepAI/nucleotide_transformer_downstream_tasks_revised`), 300-bp windows with the TSS at index 49. Positives are windows containing an annotated promoter; negatives are genomic windows not overlapping one, with no composition matching applied by the benchmark authors. Three splits are used. Training used the benchmark's training split with chr8, chr9 and chr18 removed: 26,699 sequences, 13,611 of them promoters and 392 of those with a canonical TATA box; 10% were drawn at random for validation, which sets early stopping and decision thresholds. Test AUROC is computed on the benchmark's official test split (chr20 and chr21). Grammar discrimination is computed on a separate probe: the 76 canonical-TATA promoters on chr8, chr9, chr18, chr20 and chr21 in the benchmark's promoter tables, none of which occurs in training. With 76 probe promoters GD carries appreciable sampling error, so intervals on the gain resample promoters as well as seeds.

External measurements comprised: a saturation-mutagenesis massively parallel reporter assay (MPRA) covering nine disease promoters [26]; TBP–DNA dissociation constants for 14 disease-associated promoter SNPs [27]; and lentiMPRA measurements of regulatory activity for 452,706 200-bp sequences pooled across three cell lines, split 362,163 / 45,272 / 45,271 [28] (Section II-G).

B. Models

Six architectures spanning common inductive biases were used.

- **CNN-GlobalPool:** a dilated convolutional trunk (stem of 128 filters width 15; three residual blocks at dilations 2, 4, 8; receptive field 71 bp) with a translation-invariant readout concatenating global max and mean pooling.
- **CNN-Dense:** the same trunk with a position-aware readout (max-pool by 5, flatten, dense), included to test whether the elevated null depends on the pooling rather than the trunk.
- **Transformer:** convolutional stem, stride-3 downsampling, three encoder layers (width 96, four heads) with learned absolute positional embeddings and a CLS readout.

- **NT-v2-50M / NT-v2-100M:** the pretrained Nucleotide Transformer v2 [29], fully fine-tuned, with 6-mer tokenisation of the 300-bp window.
- **NT-v2-500M (LoRA):** the 500M-parameter model with low-rank adapters [30] of rank 16 on attention query and value projections, 1.90M trainable parameters of 495M (0.38%), with activation checkpointing.

All methods share binary cross-entropy loss, AdamW (weight decay 0.01) and early stopping on validation AUROC. The convolutional and transformer models train for 12 epochs and the pretrained models for 3. Learning rates are 1×10^{-3} (CNN-GlobalPool), 1×10^{-4} (CNN-Dense), 5×10^{-4} (Transformer), 5×10^{-5} (NT-v2-50M/100M) and 3×10^{-4} (LoRA adapters), fixed across every training method so that no method received extra tuning. Batch size is 128, reduced to 64 for NT-v2-100M and 32 for the 500M model to fit 8 GB of GPU memory. CNN-Dense uses the lower rate because at 3×10^{-4} one seed in three failed to converge. Decision thresholds are chosen on the validation split to maximise the Matthews correlation coefficient. To test a design not chosen for this study, DeepSTARR [31] was re-implemented with its published layer sizes (four convolution blocks and two batch-normalised dense layers of 256 units), given a single classification output, and trained identically with the CNN-GlobalPool learning rate, with its own null and GCR, three seeds each.

C. Composition-Preserving Counterfactuals and Grammar Discrimination

The canonical TATA box was located by scanning the JASPAR 2026 TBP profile MA0108.3 in the window -35 to -25 and requiring a score at least 70% of the profile maximum. Edits are block swaps: the 8-bp block containing the element is exchanged with an equally sized block at a distant downstream position. Swapping two blocks of equal width permutes the window's bases without changing the multiset, so mononucleotide composition is preserved exactly.

Grammar discrimination is the area under the ROC curve separating the signed score changes caused by displacing the TATA block from those caused by displacing a control block of the same width. Control blocks lie in the same promoter, immediately on either side of the TATA box: they start at -49 to -45 or at -20 to -16 relative to the TSS, against -35 to -25 for the TATA box. The blocks avoid the TATA and Inr positions but not every core element: the downstream blocks span -20 to -9 and overlap the BREd position (about -23 to -17), and the upstream blocks reach -38, the edge of BREu [2]. TATA and control blocks are swapped to the same set of downstream targets, at $+51$ and beyond. GD therefore compares the TATA position with its immediate core-promoter neighbours in the same sequence, so sensitivity to edits near the core promoter in general does not raise it. The two edits do still differ in what they move: the TATA block is AT-rich and its neighbours generally are not, so a model that is sensitive to local base composition near the TSS can separate them without representing the TATA motif. GD controls for the composition of the window, not for composition at a particular position, and the null models below measure how much of a model's GD that accounts for.

Proposition (by construction). Let s be any scoring function depending on a window only through its mononucleotide composition. Both the TATA displacement and the control displacement are block swaps, which preserve that composition, so s returns the same value before and after either edit. The two sets of score changes are therefore identical — each is the zero vector — and GD is exactly 0.5.

A score above 0.5 thus excludes an explanation in terms of mononucleotide composition. It does not exclude one in terms of k-mer composition: each block swap changes the dinucleotides at the four block junctions, so scorers reading k-mers with $k \geq 2$ can depart from 0.5 (Section III-B). Nor does a score above 0.5 establish knowledge of the TATA box, which motivates the null models below.

Two further measures use edits never used in training: shift sensitivity, the mean score drop when the block is moved by at least 8 bp, standardised by each model's logit standard deviation on the test set; and TATA recognition, the AUROC separating a double substitution destroying the motif from a matched double substitution elsewhere.

D. Architecture-Matched Null Models

For each architecture we retrained from scratch on the same data with all 392 canonical-TATA promoters removed, holding every hyperparameter fixed. The resulting GD is the score that architecture obtains without any canonical-TATA promoter to learn from, and is the reference against which its trained score must be read. It is not a model without exposure to TATA-like sequence. The AT-rich negatives contain TATA-like hits at other positions (mean best TBP log-odds $+6.83$), 297 negatives carry a canonical TATA box at -30, and some remaining promoters carry sub-threshold TATA boxes. What the null removes is the association between a TATA box at -30 and the promoter label, which is the association GD is designed to detect. A stricter null for two architectures, with the canonical TATA box at -30 removed from both classes (689 sequences), gave the same reference level: GD 0.519 ± 0.061 against 0.483 ± 0.062 for the transformer and 0.804 ± 0.007 against 0.784 ± 0.032 for CNN-GlobalPool (three seeds each). The TATA boxes left in the negatives therefore do not account for the height of the null. The removal takes 1.5% of the training set, and every null model reaches test AUROC within 0.003 of its trained counterpart, so the difference in GD is not attributable to the loss of those sequences as training data.

Two further experiments test why the baselines fall short. In composition-matched retraining, the Transformer, CNN-GlobalPool and NT-v2-50M were retrained with the genomic negatives replaced by the training promoters themselves, each shuffled within consecutive 20-bp segments with a dinucleotide-preserving shuffle. GC content and its profile along the window are then identical in the two classes: GC content alone classifies at AUROC 0.500, and the TBP profile at 0.536 rather than 0.212, so the motif is no longer penalised. The matching null removes every canonical-TATA promoter

before the shuffled partners are made, and GD is measured on the same probe. In the frequency experiment, Transformers were trained on 3,000 promoters in which 0%, 1%, 3%, 10% or 16% carried a canonical TATA box at -30, three seeds each, once against 3,000 genomic negatives and once against the same promoters shuffled as above, so that rarity can be varied with and without the penalty. About 500 canonical-TATA promoters exist on the training chromosomes, so 16% of 3,000 is close to the most the benchmark allows.

Sensitivity to core elements that GCR does not protect was measured on the probe promoters as the mean absolute change in logit over all single-nucleotide substitutions in BREu and BREd (the 7 bp on either side of the TBP core) and the Inr (-2 to $+4$), divided by the same quantity at $+61$ to $+100$.

E. Grammar-Consistency Regularization

Every step draws an auxiliary batch of 32 canonical-TATA training positives x, with counterfactuals $\tilde{x}$ (TATA block displaced) and $\hat{x}$ (position-matched control block displaced). Writing z for the model logit and m for the margin, the objective is

$$\mathcal{L} = \mathrm{BCE} + \lambda_r \cdot \mathrm{mean}\ \max(0,\, m - (z(x) - z(\tilde{x}))) + \lambda_i \cdot \mathrm{mean}\ (z(x) - z(\hat{x}))^2 \quad (1)$$

with $\lambda_r = \lambda_i = m = 1$ throughout, except on lentiMPRA where λ was reduced to 0.1 (Section II-G). The ranking term asks only that displacing the TATA box lowers the promoter score; the invariance term requires that a position-matched control displacement does not. Auxiliary sequences never enter with a label, so GCR does not assert that an edited sequence is a non-promoter — the assumption that distinguishes it from a hard-negative baseline, which we include for comparison. Each GCR step evaluates the model on 96 sequences beyond the main batch (the 32 native sequences and their two counterfactuals), so the sequences processed per step rise by a factor of $(B + 96)/B$: 1.75 at the batch size of 128 used for most models, 2.5 for NT-v2-100M and 4.0 for the 500M model.

For the lentiMPRA regression the cross-entropy term is replaced by mean squared error on standardised activity, with the same two constraint terms and the same margin.

GCR-Atlas extends this to 60 non-redundant positional rules derived by scanning all 1,019 JASPAR 2026 CORE vertebrate profiles [32] for positional enrichment in training promoters.

Preserving composition under permutation is long-standing practice in sequence-significance testing [33]. Block swaps differ in applying that preservation within the single window being scored, so each comparison is against the same sequence rather than against a background ensemble.

F. Falsification Controls

Two wrong-rule controls were trained with identical loss, margin, auxiliary batch size and seeds. Decoy GCR protects a fixed block at $+71$, where no core-promoter element is positionally constrained. Scrambled-atlas GCR retains every motif, threshold and width but moves each rule's window 80 bp from where the atlas found it enriched.

G. Replication on Measured Regulatory Activity, and on Splice Sites

lentiMPRA sequences are 200 bp with a continuous activity readout. Scanning the TBP profile over the promoter subset places the modal best hit at index 69, at 3.3 times the uniform rate on the training split, consistent with a TATA box near -30 relative to a TSS at approximately index 98. The rule coordinates are taken directly from this TBP enrichment; the implied TSS position assumes the canonical -30 spacing and is not otherwise used. The dataset distributes HepG2, K562 and WTC11 measurements as separate files, which were pooled without a cell-line indicator. Models were trained by regression on all 362,163 training sequences for 6 epochs and evaluated on the 45,271 held-out sequences. Because some sequences were measured in more than one cell line, 11.9% of the unique test sequences also occur in training with another cell line's measurement; the evaluation is therefore repeated on the 39,749 test sequences absent from training and on each cell line separately (Table S10).

On lentiMPRA the constraint weights were reduced to $\lambda_r = \lambda_i = 0.1$. At the default weight of 1, a single exploratory CNN run reduced predictive correlation from 0.627 to 0.556, and a run at 0.3 reduced it to 0.518. Both runs were scored on the held-out set, so the weight of 0.1 was chosen after seeing held-out results; the cost reported in Section III-J is the smallest of the three settings tried, and the cost at the default weight was about 11%. Scored afterwards on the validation split, which no choice had used, the three weights rank in the same order: correlation 0.553 at $\lambda = 1$, 0.523 at 0.3 and 0.594 to 0.618 at 0.1, against 0.626 to 0.630 for the baseline. The same weight would therefore have been selected without reference to the test set. The null condition removes every canonical-TATA sequence from training, as for the promoter benchmark. Orientation was tested on the 1,964 held-out sequences present as exact reverse-complement pairs with two measured activities, comparing the measured difference with the difference in model score.

Splice-donor classification used the 600-bp windows of the same benchmark family, where the donor GT sits at index 301 in 51% of positives against a 5% background rate. Because every positive contains the core dinucleotide, there is no equivalent of "promoters without a TATA box"; the null instead applies a random circular shift to each window, which preserves all local sequence content while destroying absolute position.

H. Statistics

Comparisons on the MPRA resample whole promoters, because variants within a promoter are not independent, with 5,000 bootstrap replicates. Reported p-values are unadjusted; where the same contrast is tested in several architectures, the Holm-adjusted value over those architectures is given alongside. Seeds were ensembled before computing promoter-level statistics; this was specified in advance, and per-seed statistics are given in Table S2. The Transformer, CNN-GlobalPool and NT-v2-50M use five seeds for baseline, GCR and GCR-Atlas; all other configurations use three. We report

mean ± s.d. over seeds, and all tests are two-sided. Confidence intervals on the gain over null resample the 76 probe promoters and, within each condition, the seeds, so that they include both sources of variation (5,000 replicates).

Where a contrast was underpowered in a single architecture, we pooled the per-architecture differences within each bootstrap resample. This keeps the promoter clusters as the resampling unit, so the interval narrows only to the extent that architectures agree; it does not create power where the clusters provide none.

I. Training-Set Overlap in the External Validation

The MPRA windows were cut from hg38 around each TSS independently of the benchmark. Comparing coordinates afterwards, six of the nine overlap a benchmark training window, and one — *LDLR* — is the identical 300-bp interval present in the training split with a positive label. The remaining five overlap by 50–75 bp of 300. We therefore report the primary analysis with *LDLR* excluded and give the full overlap table in Table S1.

III. Results

A. The Benchmark Is Largely Solvable Without Position, and Penalises the Element

Logistic regression on GC content alone separated promoters from negatives at AUROC 0.898, and on k-mer frequencies up to $k = 5$ at 0.926. All six neural models reached AUROC 0.940–0.962 — between 0.014 and 0.036 above what composition alone achieves.

Scoring the TBP profile directly is more revealing. The best TBP hit classifies promoters at **AUROC 0.212**, far below chance: promoters average $+1.95$ in log-odds against $+6.83$ for negatives, because promoters are GC-rich (0.623 against 0.422) while the TATA box is AT-rich, and motif score correlates with window GC at -0.78. On this benchmark a model that uses the TATA box positively is penalised for it (Table S4).

B. Grammar Discrimination Is Composition-Blind, as Stated

Putting fitted composition models through the identical measurement confirms the proposition. GC content and a mononucleotide model both gave $\mathrm{GD} = 0.500000$ with maximum absolute score change 0.00×10^{0}, despite AUROC 0.898. Models using k-mers up to $k = 5$ gave GD between 0.471 and 0.530 at AUROC up to 0.926 (Table S3). This departure is expected, because block swaps change the dinucleotides at the block junctions; GD is blind to mononucleotide composition, not to k-mer composition.

C. Positional Scores Have No Universal Chance Level

Null models differ by more than 0.3 between architectures of equal accuracy (Fig. 1a, Table I). The transformer's null is 0.483; the global-pooling CNN's is 0.784 at almost identical test AUROC (0.942 against 0.944). Replacing global pooling with a position-aware dense readout changed the null only from 0.784 to 0.801, so the elevation follows the convolutional trunk rather than the readout. DeepSTARR, a published convolutional design trained identically, has the highest null of all, 0.844 (Table S11).

D. Accurate Models Learn Only Part of the Positional Rule

Against their own nulls, the six baselines gain between $+0.010$ and $+0.163$, which is 2% to 69% of the room above the null (Fig. 1b, Table I). This includes the largest model tested: at 495M parameters and test AUROC 0.962, the highest in the study, the gain is $+0.140$ (41%), inside the range set by models an order of magnitude smaller. With probe promoters and seeds resampled together, the gain is clearly above zero for CNN-GlobalPool, CNN-Dense, NT-v2-100M and NT-v2-500M, marginal for the Transformer (95% CI -0.018 to $+0.254$) and absent for NT-v2-50M (-0.126 to $+0.154$). The baselines therefore learn part of the positional rule, the convolutional models most, but none captures more than 69% of the room above its null, against 90% to 99% for every architecture under GCR. The tuning curves show the same thing (Fig. 2): the CNN-GlobalPool baseline peaks at the native position, but so does its null. DeepSTARR falls in the same range: GD 0.925 against its null of 0.844, a gain of $+0.082$ (95% CI $+0.030$ to $+0.143$) and 52% of the room (Table S11).

E. Removing the Composition Confound Does Not Close the Gap

With composition-matched negatives, the baselines still separated promoters from their shuffled counterparts (AUROC 0.88 to 0.93 on a matched test set), but gained little more over their matched nulls: $+0.082$ for the Transformer (25% of the room above the null, against 23% on the original benchmark), $+0.105$ for CNN-GlobalPool (30%, against 56%) and $+0.045$ for NT-v2-50M (13%, against 2%). For the Transformer and NT-v2-50M the 95% intervals on these gains include zero (Table S11). The matched nulls are themselves higher than the original ones for two of the three architectures (0.665 against 0.483 for the Transformer, 0.641 against 0.525 for NT-v2-50M), so matching changes what a model without TATA knowledge can achieve as well as what the baseline learns. GCR on the same data reached GD 0.996, 0.943 and 0.987, or 84% to 99% of the room. Removing the penalty on the motif therefore did not close the gap, and the anti-predictive motif is not what chiefly limits the baselines.

Rarity contributes, but does not account for the gap either (Table S12). With composition-matched negatives, the Transformer's GD rose with the share of TATA promoters, from 0.527 with none to 0.718 at 10%, but no further at 16% (0.710), whereas GCR reached 0.971 with 1%. With genomic negatives, where the motif is penalised, GD stayed lower throughout (0.434 to 0.584). More examples of the element, within the counts this benchmark provides, did not bring the baseline close to the constrained model.

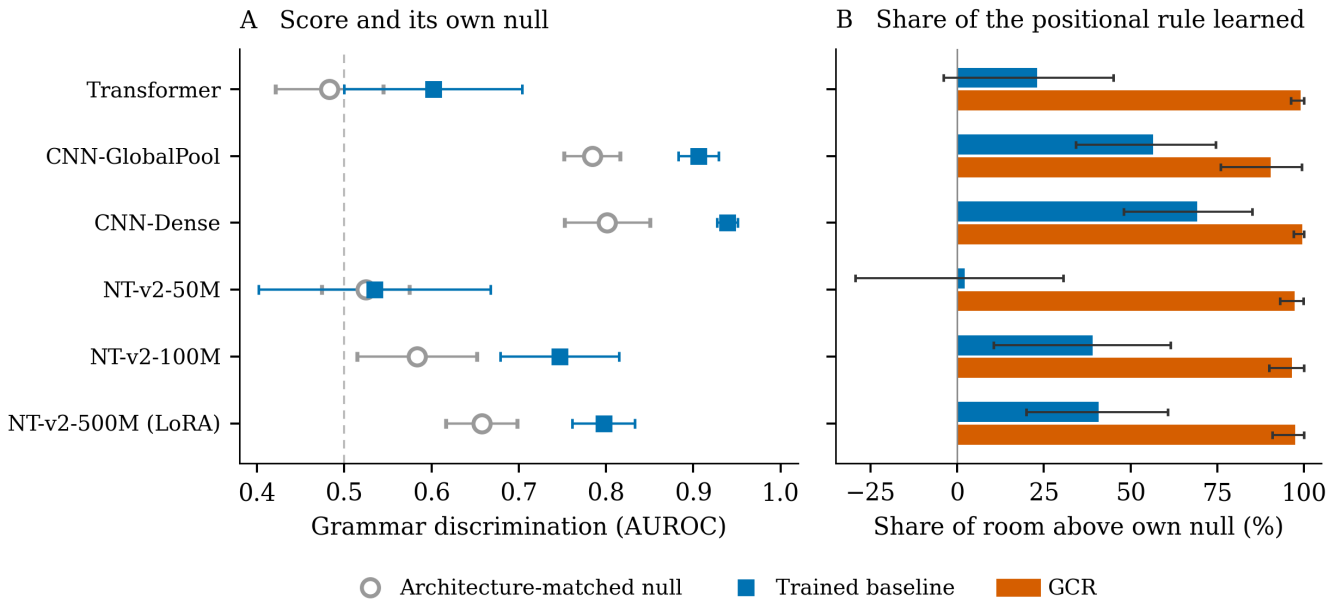


Fig. 1. Accurate models capture only part of the room above their own architecture-matched nulls. (a) Grammar discrimination for each architecture (filled squares) beside the null model obtained by retraining that architecture with every canonical-TATA promoter removed (open circles). Markers are means over seeds, bars one standard deviation; the dashed line marks 0.5. (b) Share of the room above each model's own null, (GD − null) / (1 − null), for the trained baseline and for GCR, with 95% intervals that resample probe promoters and seeds.

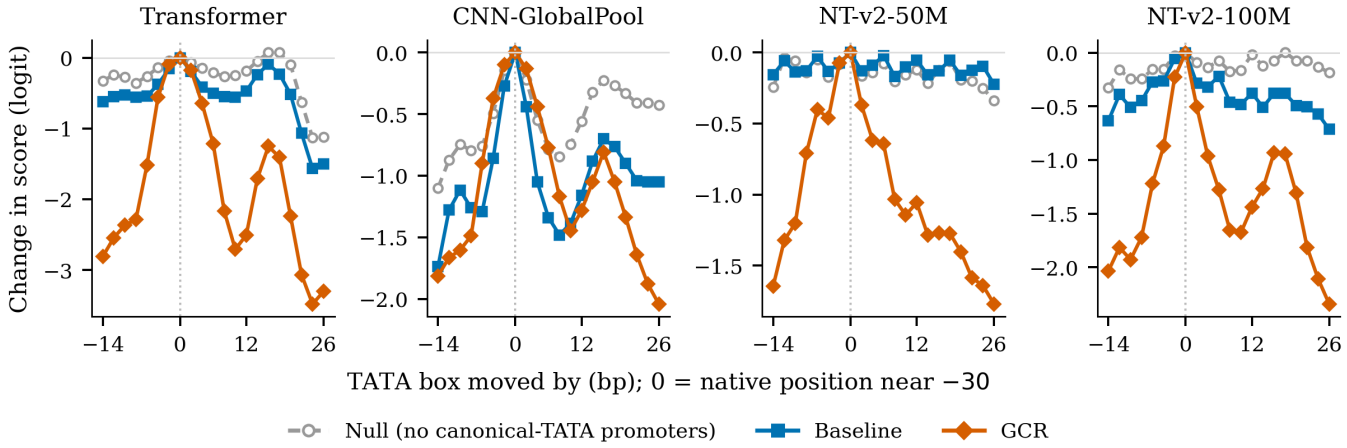


Fig. 2. Positional tuning, against each architecture's null. Mean change in promoter score when the TATA box of held-out promoters is displaced by −14 to +26 bp, where 0 is the native position near −30, for the null model (grey, open circles), the trained baseline and GCR. None of these shifts was used in training, which used only displacements beyond +51 bp. Shown for four of the six architectures; CNN-Dense and NT-v2-500M are summarised in Table I.

TABLE I
EACH MODEL AGAINST ITS OWN ARCHITECTURE-MATCHED NULL

Model	Seeds	Test AUROC	Baseline GD	Null GD	Gain [95% CI]	Share	GCR GD	GCR share
Transformer	5 / 3	0.942	0.602 ± 0.102	0.483 ± 0.062	+0.119 [−0.018, +0.254]	23%	0.995 ± 0.006	99%
CNN-GlobalPool	5 / 3	0.944	0.906 ± 0.023	0.784 ± 0.032	+0.122 [+0.060, +0.187]	56%	0.979 ± 0.009	90%
CNN-Dense	3 / 3	0.948	0.939 ± 0.012	0.801 ± 0.049	+0.138 [+0.066, +0.213]	69%	0.999 ± 0.001	99%
NT-v2-50M	5 / 3	0.940	0.535 ± 0.133	0.525 ± 0.050	+0.010 [−0.126, +0.154]	2%	0.987 ± 0.009	97%
NT-v2-100M	3 / 3	0.951	0.747 ± 0.068	0.584 ± 0.069	+0.163 [+0.037, +0.292]	39%	0.985 ± 0.009	96%
NT-v2-500M (LoRA)	3 / 3	0.962	0.797 ± 0.036	0.657 ± 0.041	+0.140 [+0.061, +0.224]	41%	0.991 ± 0.006	98%

GD is mean ± s.d. over seeds, measured on 76 held-out canonical-TATA promoters. Seeds are given as baseline / null; GCR uses the baseline seeds. Gain is baseline minus null GD; its interval resamples the probe promoters and, within each condition, the seeds (5,000 replicates). Share is the gain as a fraction of the room above the null, (GD − null) / (1 − null).

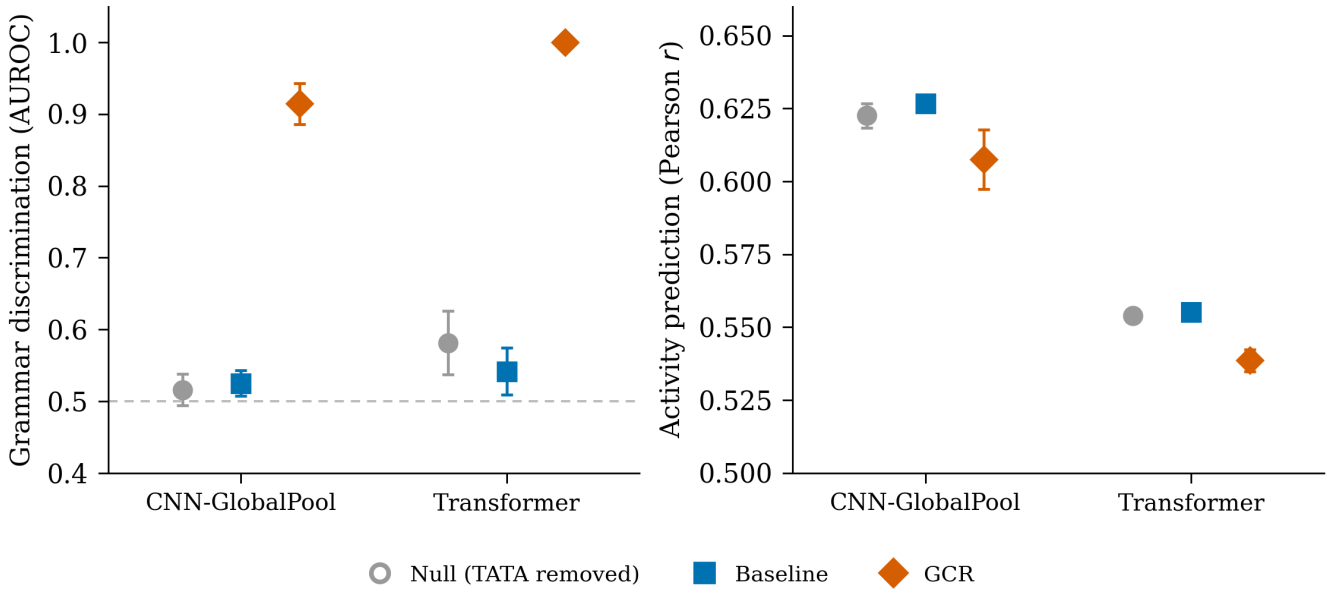


Fig. 3. The audit replicates on measured regulatory activity. Grammar discrimination (left) and activity prediction (right) on 45,271 held-out lentiMPRA sequences, for the null, the trained baseline and GCR, in two architectures.

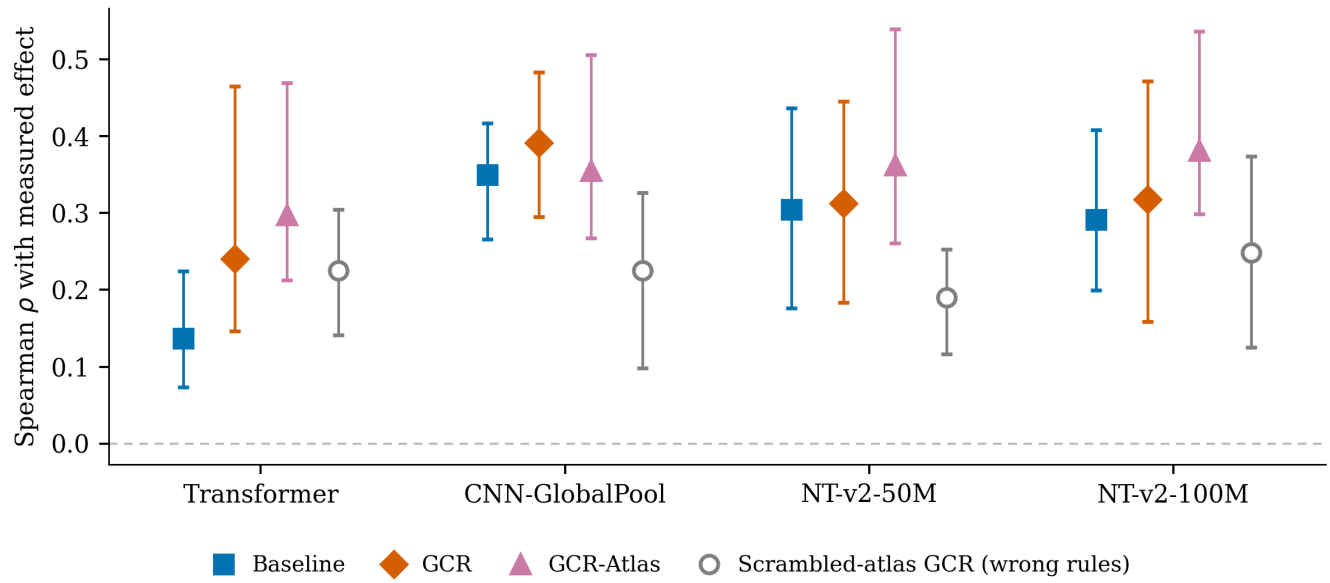


Fig. 4. Agreement with measured variant effects, primary analysis. Spearman correlation between seed-ensembled predictions and MPRA-measured effects for 475 variants in eight promoters (*LDLR* excluded), with 95% promoter-level bootstrap intervals for each condition alone; the paired comparisons between conditions are in Table S5.

F. GCR Restores Positional Use at Little Accuracy Cost

GCR raised the gain over null substantially in all six architectures of Table I, to 90% to 99% of the room above the null. Test AUROC changed by at most 0.006 in either direction: it fell by 0.003 to 0.006 in four architectures, rose by 0.006 in NT-v2-50M and was unchanged in NT-v2-100M. Four of these differences exceed the corresponding seed standard deviation, so the cost on this task is small but not identically zero. The result does not depend on the constraint's settings: varying λ_r and λ_i threefold and m twofold in each direction for the Transformer gave GD of 0.983 to 0.996 and test AUROC of 0.936 to 0.944 (Table S9).

It did not work for DeepSTARR: GD fell from 0.925 to 0.806 and test AUROC from 0.944 to 0.929. GCR scores a sequence and its counterfactuals in separate forward passes, and DeepSTARR alone applies batch normalisation to its dense features, which re-centres each pass separately; one joint pass raised GD to 0.865 on average, but unreliably (0.744 to 0.988 across seeds). We did not tune the constraint for this architecture; GCR as specified is not architecture-agnostic.

Grammar discrimination is, however, measured with the

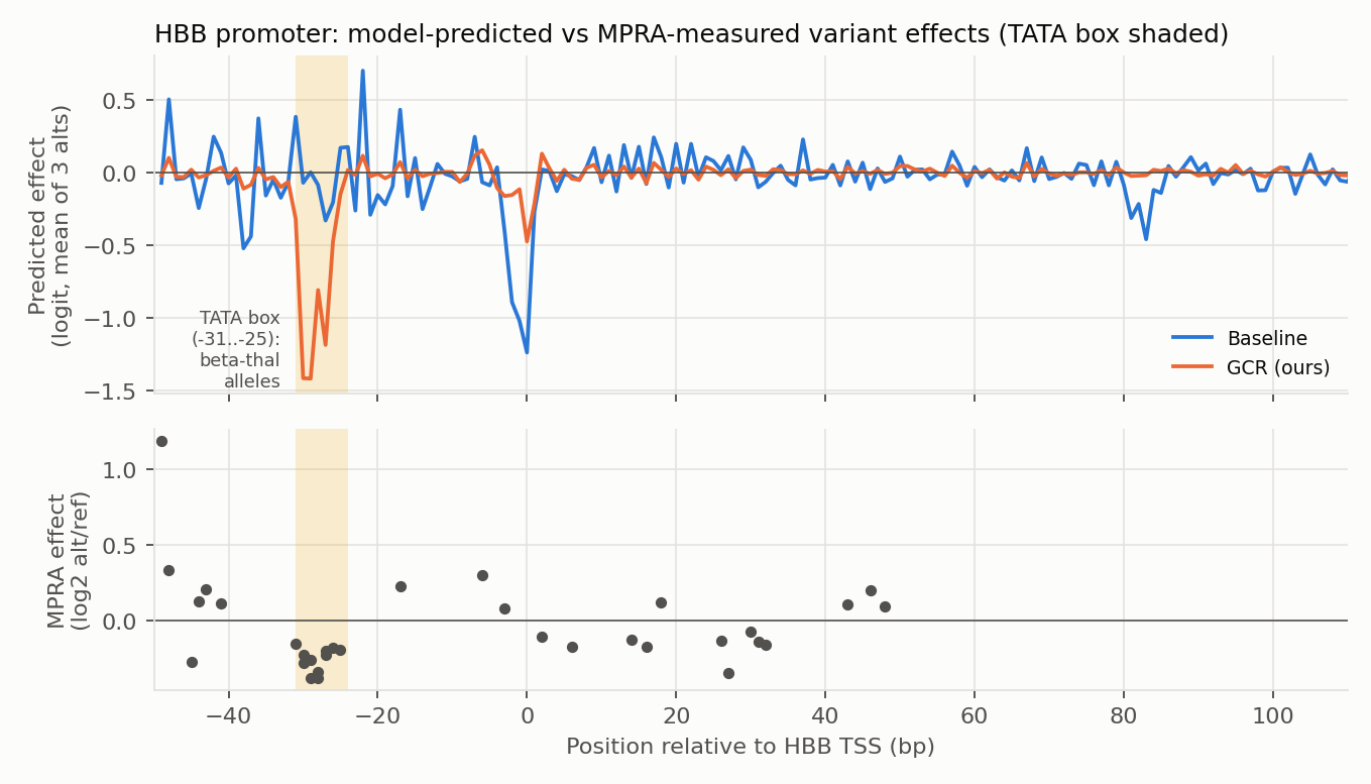


Fig. 5. Variant effects across the *HBB* promoter. Top: predicted effect of each substitution along the window for a baseline and a GCR model, with the TATA box (-31 to -25), where β-thalassemia alleles lie, shaded. Bottom: MPRA-measured effects for the same positions.

same family of edits that GCR is trained on, so a rise in GD alone would be close to circular. The evidence that the constraint produces positional behaviour rather than edit-specific memorisation comes from measurements outside that family. The response to displacement is sharply peaked at the native position across shifts of -14 to $+26$ bp, none of which was used in training, where training used only displacements beyond $+51$ bp (Fig. 2). Implanting TATAAAA into held-out promoters lacking a canonical element — an insertion, not a swap, and never performed during training — produced a response peaking within 2 bp of -30 in 21 of 21 GCR models, against 7 of 21 baselines and 0 of 15 null models (all architectures except NT-v2-500M, which was not tested). The external MPRA, TBP-binding and lentiMPRA results below involve no block swap at all.

G. Comparison With Alternative Training Interventions

Two simpler interventions were trained under the same budget, seeds and auxiliary batch size. A hard-negative baseline labels the displaced counterfactual as a non-promoter. Random augmentation applies the same block swaps as additional training sequences without any constraint linking a sequence to its edited form.

Random augmentation did not raise grammar discrimination: 0.577 for the Transformer and 0.926 for CNN-GlobalPool, against baselines of 0.602 and 0.906. Exposure to edited sequences is therefore not what produces the effect; the ordering constraint is.

The hard-negative baseline did reach comparable grammar discrimination — 0.984, 0.984 and 0.977 for the Transformer, CNN-GlobalPool and NT-v2-50M — but at a larger cost in test accuracy: AUROC fell from 0.942 to 0.920, 0.944 to 0.932 and 0.940 to 0.936 respectively, against 0.939, 0.940 and 0.946 under GCR. Its agreement with measured TBP binding was mixed (Section III-H). The ordering constraint is thus the cheaper of two routes to similar positional behaviour on this task, and we do not claim it is the only one.

H. Ranking of Disease-Causing Promoter Mutations

β-Thalassemia arises from single substitutions in the *HBB* TATA box. Scoring all 897 possible single-nucleotide variants of the *HBB* promoter window and asking where the four known pathogenic TATA-box positions rank gives a direct test of whether a model treats the element as load-bearing. Baselines placed them near the middle of the distribution — at the 42.5th percentile of damage for the Transformer and the 57.2nd for the CNN, where the 50th percentile is no discrimination. GCR moved them to the 92.6th and 94.4th percentiles respectively (Fig. 5). The two pretrained models already ranked them highly without the constraint, at the 68.6th (NT-v2-50M) and 86.3rd (NT-v2-100M) percentiles, and GCR raised these to the 89.2nd and 90.8th; the large gains are confined to the two models trained from scratch. Two qualifications apply. Scoring the TBP profile alone places the same variants at the 98.8th percentile, above every model, so GCR brings the models toward what a hand-coded profile

already encodes rather than beyond it. And the *HBB* window shares 60 bp, including its TATA box, with a training window, where that sequence appears in reverse orientation and not at -30 (Table S1).

Agreement with measured TBP binding points the same way, with a caveat. For 14 disease-associated promoter SNPs with measured TBP–DNA dissociation constants [27], Spearman correlation between predicted and measured effect rose from $+0.009$ to $+0.412$ for the Transformer and from $+0.084$ to $+0.464$ for NT-v2-50M, where the baselines were essentially uncorrelated. Where a baseline already correlated, the change was small (CNN-GlobalPool $+0.400$ to $+0.456$; NT-v2-100M $+0.411$ to $+0.489$). The hard-negative baseline gave mixed results: $+0.458$ for the Transformer, higher than GCR's $+0.412$, but $+0.242$ for CNN-GlobalPool and $+0.251$ for NT-v2-50M, below GCR's $+0.456$ and $+0.464$. The caveat is that scoring the TBP profile directly attains $+0.656$ on this set, higher than any trained model. With $n = 14$ and half of those variants in *HBB*, these correlations are indicative only, and on this particular quantity the hand-coded profile remains the better predictor.

The corresponding test for an activating mutation is negative and we report it as such. The *TERT* promoter mutation C228T creates a de novo ETS site and is the most common non-coding mutation in several cancers [34]. Ranked among all single-nucleotide variants of the *TERT* window by predicted activation, it reached the 44th to 47th percentile under GCR-Atlas in three architectures and the 75th in CNN-GlobalPool, against the 32nd to 53rd for baselines. One architecture of four is not evidence of a general ability to recognise gain-of-function variants, and we make no such claim.

I. External Agreement Depends on the Rules Being Correct

Neither wrong-rule control learned the TATA rule. Grammar discrimination fell below the corresponding baseline under both: to 0.326 under decoy GCR (Transformer baseline 0.602), and to 0.434, 0.470, 0.430 and 0.613 under scrambled-atlas GCR for the Transformer, CNN-GlobalPool, NT-v2-50M and NT-v2-100M (baselines 0.602, 0.906, 0.535 and 0.747). Decoy GCR also cost accuracy (AUROC 0.914 against 0.942; Table S6).

On the MPRA with *LDLR* excluded, GCR-Atlas agreed with the measured variant effects more closely than the baseline in all four architectures tested, but the difference in Spearman ρ was significant only for the Transformer: $+0.161$ (95% CI $[+0.054, +0.313]$, $p = 0.006$; Holm-adjusted over the four architectures, $p = 0.026$), against $+0.005$ ($p = 0.81$) for CNN-GlobalPool, $+0.058$ ($p = 0.42$) for NT-v2-50M and $+0.090$ ($p = 0.20$) for NT-v2-100M (Fig. 4). The single-rule constraint was significant at the unadjusted level only in CNN-GlobalPool ($+0.042$, $p = 0.03$), which does not survive the same adjustment (Holm $p = 0.12$; Table S5). We therefore do not claim that either constraint improves agreement with measured effects in general; after correction, one comparison of eight does. In that architecture the scrambled atlas also exceeded the baseline ($\rho = 0.225$ against 0.137, and 0.298 for GCR-Atlas), so part of the Transformer's improvement may be a generic effect of the constraint rather than of the rules being correct.

The comparison with the scrambled atlas is more consistent. Correct rules exceeded displaced rules in all four architectures, with $p < 0.05$ in three (CNN-GlobalPool $+0.130$, NT-v2-50M $+0.172$, NT-v2-100M $+0.133$) and not in the Transformer ($+0.073$, $p = 0.19$); pooling the per-architecture differences within each bootstrap resample gave $+0.139$ $[+0.034, +0.288]$, $p = 0.01$ (Table S5). The two comparisons do not say the same thing. In CNN-GlobalPool, where GCR-Atlas did not differ from the baseline, the contrast with the scrambled atlas arises because displaced rules lowered agreement, not because correct rules raised it. What the external data support is therefore that the rules must be correct for a constraint of this kind to be harmless or helpful, and that the correct rules improve on an unconstrained model in one architecture of four.

J. The Audit Replicates on an Independent Dataset of Measured Activity

On lentiMPRA, baselines again sat at their nulls (Fig. 3): GD 0.525 against a null of 0.516 for the CNN, and 0.542 against 0.581 for the transformer — the latter below its null. GCR raised GD to 0.914 and 1.000 respectively.

This came at a cost. At $\lambda = 0.1$, predictive correlation fell from 0.627 to 0.608 (CNN) and from 0.555 to 0.539 (transformer), about 3% relative, consistent across all three seeds in each condition with non-overlapping ranges; at the default weight the cost was about 11% (Section II-G). Restricted to the 39,749 test sequences not present in training, every correlation was lower by about 0.045, but the comparison was unchanged: 0.582 to 0.563 (CNN) and 0.504 to 0.489 (transformer), again with non-overlapping seed ranges. The cost appeared in each cell line separately, at 1.6% to 5.8% (Table S10). On this dataset GCR trades predictive accuracy for positional grammar; it does not improve prediction of measured activity.

IV. Discussion

High benchmark accuracy does not establish that a DNA model has learned the mechanism it is used to interpret. On a task that composition alone nearly solves, six accurate promoter models, including a 495M-parameter genomic language model, capture at most 69% of the room above models that never saw a canonical-TATA promoter, and one captures almost none.

The shortfall is not explained by the composition confound. The TATA box is anti-predictive on this benchmark, but with composition-matched negatives, where it no longer is, the baselines still captured only 13% to 30% of the room above their nulls. Rarity contributes: a rule carried by 2.9% of positives contributes little to the training loss, and without the penalty the Transformer's GD rose as the rule became more common. But it stopped rising at about 0.72, far below what the constraint reached with 1% of promoters, so neither the penalty nor rarity alone explains the gap. Models trained on a task in which the element shapes the output, such as

base-resolution transcription initiation [16], [17], do learn its position; for classifiers and activity models, an explicit constraint changed it here.

Two methodological points generalise beyond this element. First, positional-interpretability scores are not comparable across architectures unless each is referred to its own null; a convolutional model scored 0.80 having never seen a canonical-TATA promoter. Reports of positional knowledge in sequence models should be read with this in mind [35]. Second, a measure built from composition-preserving edits can be shown blind to mononucleotide composition by construction, which converts it from a heuristic into a bounded measurement.

A. Limitations

The accuracy cost is real and task-dependent. On the promoter benchmark it is at most 0.006 of AUROC and falls in four of six architectures; on lentiMPRA it is about 3% of predictive correlation at the weight used and about 11% at the default weight, and the weight was chosen after seeing held-out results, although the validation split selects the same weight. We report all of these, and make no claim that the constraint is free.

The invariance term penalises sensitivity at the control positions, and GCR reduced the mean absolute effect of control displacements by a factor of 2.0 to 6.5. Because the downstream control blocks overlap BREd, the constraint can suppress sensitivity there, and measured directly it sometimes did (Table S13). Relative to distal sequence, sensitivity at BREd fell under GCR by 15% to 37% in the Transformer, CNN-Dense and DeepSTARR, and at the Inr by 24% to 28% in CNN-Dense and DeepSTARR; neither fell in CNN-GlobalPool, and all three rose in the NT models. In *HBB*, GCR also more than halved the Transformer's predicted effect of substitutions at the TSS (Fig. 5). Null models are about as sensitive at these positions as the baselines, so that sensitivity does not by itself show knowledge of the elements. A constraint that protects one element can cost sensitivity at its neighbours, which argues for protecting the known core elements together.

Only models trained here were audited. DeepSTARR shows that the audit extends to a published design, but only as retrained on this benchmark. Widely used pretrained models such as Enformer [13] cannot practically be retrained without the element, so their GD could not be referred to an architecture-matched null, which Section III-C shows is required to interpret it.

Transfer to a structurally different element is weak. On splice-donor classification, baselines again scored at their null (0.608 against 0.621), but GCR gained only $+0.082$ over null while costing 7 points of AUROC (0.854 to 0.785; Table S7). For the same element, GCR transferred to a second dataset with a different sequence length and a regression objective; it does not transfer well to an element whose constraint differs in kind.

GCR models respond more strongly than baselines when the motif is reverse-complemented in place, but this is not orientation knowledge: on 1,964 lentiMPRA sequences measured in both orientations they predicted the orientation effect no better than baselines ($r = 0.252$ against 0.262; Table S8).

The external MPRA covers nine promoters, of which six overlap training windows and one is identical to a training sequence. Excluding that promoter leaves eight clusters, which is a small basis for an external claim. Removing all six overlapping promoters leaves three, and on those three the difference is $+0.002$ ($p = 0.51$): the effect is absent. Three clusters cannot support either conclusion, but we report it rather than resting on the $n = 8$ analysis alone; this is the principal limitation of the external validation.

Positional enrichment in the atlas does not by itself prove function; the profiles are computational predictions rather than measured occupancy, and maps of experimentally supported binding sites [36] would constrain the rule set more tightly. Ten of the 60 rules fail a stricter null yet were retained in training. The TBP binding set is small ($n = 14$, half in *HBB*). The 300-bp windows exclude distal elements. The MPRA measures episomal reporters in one cell type.

V. Conclusion

Accurate sequence models can learn only part of the positional rule for an element whose disruption causes disease, and an apparent positional score can be an artefact of architecture rather than knowledge of the element. Composition-preserving edits make the first measurable without confounding by base composition, and architecture-matched nulls make the second visible. Removing the composition confound from training did not close the gap; a ranking constraint over the same edits did in six of seven architectures, capturing 84% to 99% of the room above the null both on the original task, where the element is penalised, and on a composition-matched version, where it is not. Its external benefit is narrower: correct rules consistently outperform displaced ones, but after correction for multiple testing improve on an unconstrained model in one architecture of four. Its costs are small, task-dependent, and reported here rather than assumed away.

Data Availability

All input data are public: the Nucleotide Transformer benchmark (Hugging Face `InstaDeepAI/nucleotide_transformer_downstream_tasks_revised`); JASPAR 2026 CORE vertebrate profiles; the saturation-mutagenesis MPRA (`gonzalobenegas/sat_mut_mpra`); lentiMPRA activity measurements (`zhangtaolab/human_lentiMPRA`); and TBP dissociation constants transcribed from [27].

Code Availability

All analysis code and result files are available at https://github.com/Maheshbonthada/promoter-grammar-gcr, and trained checkpoints at https://huggingface.co/Sravankumarbonthada/promoter-grammar-gcr. Both are public and require no registration. Every model reported here is reproducible from the deposited code and the seeds recorded in the result files; checkpoints too large to deposit are available from the author on request.

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Claude (Anthropic) [37], a large language model, was used in every section of this work to write and refactor analysis code, to draft and edit the text, and to assist with literature search. The experiments were designed and run by the author, every reported value was checked against the deposited result files, and the author takes full responsibility for the content.

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Supplementary Material

Accurate Promoter Models Only Partly Learn TATA-Box Position: An Audit With Architecture-Matched Null Models

Sravan Kumar Bonthada

All tables are generated from the deposited result files. Mean ± s.d. is over training seeds.

TABLE S1
TRAINING-SET OVERLAP OF THE SATURATION-MUTAGENESIS MPRA WINDOWS

Promoter	Chr.	Window (hg38)	Overlap (bp)	Primary analysis
F9	chrX	139,530,689–139,530,989	0 / 300	yes
GP1BA	chr22	19,723,489–19,723,789	0 / 300	yes
HBB	chr11	5,226,820–5,227,120	60 / 300	yes
HBG1	chr11	5,249,606–5,249,906	60 / 300	yes
HNF4A	chr20	44,355,649–44,355,949	0 / 300	yes
LDLR	chr19	11,089,413–11,089,713	300 / 300	no (excluded)
MSMB	chr10	46,046,018–46,046,318	60 / 300	yes
PKLR	chr1	155,301,187–155,301,487	50 / 300	yes
TERT	chr5	1,294,817–1,295,117	75 / 300	yes

Windows were cut around each TSS independently of the benchmark; overlap was found by comparing coordinates afterwards. *LDLR* is the same 300-bp interval as a training sequence and is excluded from the primary analysis.

TABLE S2
PER-SEED STATISTICS (MEAN ± S.D. OVER SEEDS)

Model	Condition	Seeds	Test AUROC	Grammar discrimination
Transformer	Null	3	0.941 ± 0.001	0.483 ± 0.062
	Baseline	5	0.942 ± 0.003	0.602 ± 0.102
	GCR	5	0.938 ± 0.002	0.995 ± 0.006
CNN-GlobalPool	Null	3	0.943 ± 0.002	0.784 ± 0.032
	Baseline	5	0.944 ± 0.002	0.906 ± 0.023
	GCR	5	0.940 ± 0.001	0.979 ± 0.009
CNN-Dense	Null	3	0.946 ± 0.001	0.801 ± 0.049
	Baseline	3	0.948 ± 0.001	0.939 ± 0.012
	GCR	3	0.943 ± 0.001	0.999 ± 0.001
NT-v2-50M	Null	3	0.939 ± 0.002	0.525 ± 0.050
	Baseline	5	0.940 ± 0.002	0.535 ± 0.133
	GCR	5	0.946 ± 0.002	0.987 ± 0.009
NT-v2-100M	Null	3	0.950 ± 0.001	0.584 ± 0.069
	Baseline	3	0.951 ± 0.001	0.747 ± 0.068
	GCR	3	0.952 ± 0.003	0.985 ± 0.009
NT-v2-500M (LoRA)	Null	3	0.960 ± 0.001	0.657 ± 0.041
	Baseline	3	0.962 ± 0.002	0.797 ± 0.036
	GCR	3	0.959 ± 0.001	0.991 ± 0.006

TABLE S3
COMPOSITION-ONLY SCORERS PUT THROUGH THE IDENTICAL MEASUREMENT

Scorer	Test AUROC	Grammar discrimination
GC content	0.898	0.500000
k-mers, $k \leq 1$	0.898	0.500000
k-mers, $k \leq 2$	0.909	0.526316
k-mers, $k \leq 3$	0.916	0.471260
k-mers, $k \leq 4$	0.925	0.483206
k-mers, $k \leq 5$	0.926	0.530125

Mononucleotide scorers obtain exactly 0.5, as the Proposition requires. Higher-order k-mer scorers are not covered by the Proposition and deviate slightly from 0.5.

TABLE S4
HAND-CODED SCORERS ON PROMOTER CLASSIFICATION

Scorer	Test AUROC
GC	0.898
PWM anywhere	0.212
PWM at -30	0.285
GC + PWM at -30	0.898

The TBP profile scores below chance because promoters are GC-rich while the TATA box is AT-rich.

TABLE S5
EXTERNAL AGREEMENT WITH MPRA-MEASURED VARIANT EFFECTS, EVERY ARCHITECTURE (*LDLR* EXCLUDED, EIGHT PROMOTER CLUSTERS)

Architecture	Contrast	$\Delta\rho$	95% CI	p	Holm p
Transformer	GCR-Atlas − baseline	$+0.161$	$[+0.054, +0.313]$	0.006	0.026
	GCR − baseline	$+0.104$	$[-0.009, +0.286]$	0.066	
	GCR-Atlas − scrambled	$+0.073$	$[-0.044, +0.251]$	0.189	
CNN-GlobalPool	GCR-Atlas − baseline	$+0.005$	$[-0.054, +0.111]$	0.808	0.846
	GCR − baseline	$+0.042$	$[+0.006, +0.090]$	0.030	
	GCR-Atlas − scrambled	$+0.130$	$[+0.012, +0.316]$	0.035	
NT-v2-50M	GCR-Atlas − baseline	$+0.058$	$[-0.068, +0.233]$	0.423	0.846
	GCR − baseline	$+0.008$	$[-0.030, +0.055]$	0.716	
	GCR-Atlas − scrambled	$+0.172$	$[+0.054, +0.354]$	0.009	
NT-v2-100M	GCR-Atlas − baseline	$+0.090$	$[-0.049, +0.259]$	0.203	0.608
	GCR − baseline	$+0.026$	$[-0.069, +0.095]$	0.514	
	GCR-Atlas − scrambled	$+0.133$	$[+0.001, +0.320]$	0.048	

Difference in Spearman ρ between seed-ensembled predictions and measured effects; promoter-level bootstrap, 5,000 replicates. Holm adjustment is over the four GCR-Atlas − baseline tests.

TABLE S6
GRAMMAR DISCRIMINATION UNDER THE WRONG-RULE CONTROLS

Architecture	Baseline	Decoy GCR	Scrambled-atlas GCR
Transformer	0.602	0.326	0.434
CNN-GlobalPool	0.906	—	0.470
NT-v2-50M	0.535	—	0.430
NT-v2-100M	0.747	—	0.613

Decoy GCR was trained for the Transformer only. Both wrong-rule controls fall below the corresponding baseline.

TABLE S7
SPLICE-DONOR CLASSIFICATION (CNN-GLOBALPOOL)

Condition	Test AUROC	Grammar discrimination
Null (position randomised)	0.850 ± 0.009	0.621 ± 0.013
Baseline	0.854 ± 0.006	0.608 ± 0.036
GCR	0.785 ± 0.007	0.703 ± 0.021

The null applies a random circular shift to each window, preserving local sequence content while destroying absolute position.

TABLE S8
PREDICTION OF MEASURED ORIENTATION EFFECTS (1,964 REVERSE-COMPLEMENT PAIRS, LENTIMPRA)

Model	Seeds	Pearson r
Baseline	3	0.262 ± 0.006
GCR	3	0.252 ± 0.007

GCR does not predict the measured orientation effect better than the baseline.

TABLE S9
SENSITIVITY OF GCR TO ITS HYPERPARAMETERS (TRANSFORMER)

λ_r	λ_i	m	Seeds	Test AUROC	Grammar discrimination	*HBB* pct.
0.3	1	1	3	0.942 ± 0.001	0.995 ± 0.006	90.6
1	0.3	1	3	0.944 ± 0.001	0.996 ± 0.006	89.7
1	1	0.5	3	0.944 ± 0.002	0.991 ± 0.011	90.1
1	1	1	3	0.941 ± 0.003	0.983 ± 0.012	98.8
1	1	2	3	0.938 ± 0.002	0.991 ± 0.006	93.4
1	3	1	3	0.936 ± 0.003	0.987 ± 0.011	95.6
3	1	1	3	0.938 ± 0.005	0.996 ± 0.006	91.2

λ_r and λ_i were varied threefold and m twofold in each direction, with the others held at 1. The configuration used throughout the paper is $\lambda_r = \lambda_i = m = 1$.

TABLE S10
LENTIMPRA ACTIVITY PREDICTION BY TEST SUBSET (PEARSON r, $\lambda = 0.1$)

Architecture	Test subset	n	Baseline	GCR	Relative cost
CNN-GlobalPool	All test sequences	45,271	0.627 ± 0.002	0.608 ± 0.010	3.1%
	Not present in training	39,749	0.582 ± 0.001	0.563 ± 0.010	3.3%
	HepG2	15,379	0.654 ± 0.004	0.616 ± 0.016	5.8%
	K562	23,056	0.678 ± 0.005	0.659 ± 0.015	2.8%
	WTC11	6,836	0.617 ± 0.003	0.607 ± 0.010	1.7%
Transformer	All test sequences	45,271	0.555 ± 0.001	0.539 ± 0.004	3.0%
	Not present in training	39,749	0.504 ± 0.000	0.489 ± 0.003	3.0%
	HepG2	15,379	0.550 ± 0.002	0.524 ± 0.005	4.8%
	K562	23,056	0.622 ± 0.003	0.612 ± 0.009	1.6%
	WTC11	6,836	0.563 ± 0.002	0.549 ± 0.007	2.4%

The dataset as distributed holds HepG2, K562 and WTC11 measurements as separate files, which were pooled for training. 11.9% of unique test sequences also occur in training, measured in another cell line; the second row excludes them.

TABLE S11
COMPOSITION-MATCHED RETRAINING AND A PUBLISHED ARCHITECTURE, SCORED AS IN TABLE I

Setting	Architecture	Seeds	Null GD	Baseline GD	Gain [95% CI]	Share [95% CI]	GCR share
Composition-matched	Transformer	3 / 3	0.665 ± 0.022	0.748 ± 0.072	+0.082 [−0.018, +0.195]	25% [−6, 55]	99%
	CNN-GlobalPool	3 / 3	0.652 ± 0.036	0.758 ± 0.038	+0.105 [+0.026, +0.189]	30% [9, 49]	84%
	NT-v2-50M	3 / 3	0.640 ± 0.036	0.686 ± 0.048	+0.045 [−0.047, +0.130]	13% [−15, 34]	96%
Published design	DeepSTARR	3 / 3	0.844 ± 0.037	0.925 ± 0.011	+0.082 [+0.030, +0.143]	52% [25, 73]	−24%

Composition-matched: negatives are the training promoters, each shuffled within 20-bp segments with a dinucleotide-preserving shuffle; the null removes canonical-TATA promoters before their shuffled partners are made. Published design: DeepSTARR (de Almeida et al., 2022), re-implemented and trained on the original benchmark, null as in Table I. Intervals resample the 76 probe promoters and, within each condition, the seeds.

TABLE S12
GRAMMAR DISCRIMINATION AGAINST THE NUMBER OF CANONICAL-TATA PROMOTERS IN TRAINING (TRANSFORMER)

TATA promoters	Genomic: baseline	Genomic: GCR	Matched: baseline	Matched: GCR
0 (0%)	0.434 ± 0.030	—	0.527 ± 0.052	—
30 (1%)	0.356 ± 0.042	0.974 ± 0.014	0.595 ± 0.042	0.971 ± 0.007
90 (3%)	0.414 ± 0.158	0.987 ± 0.001	0.612 ± 0.033	0.978 ± 0.016
300 (10%)	0.543 ± 0.050	0.991 ± 0.006	0.718 ± 0.007	0.982 ± 0.007
480 (16%)	0.584 ± 0.076	0.995 ± 0.006	0.710 ± 0.109	1.000 ± 0.000

Each training set has 3,000 promoters and 3,000 negatives; three seeds per cell. Genomic negatives are background windows, against which the TATA motif is penalised; matched negatives are the sampled promoters shuffled as in Table S11. About 500 canonical-TATA promoters exist on the training chromosomes, so 480 is near the ceiling and the count cannot be raised further on this benchmark. GCR is not defined without TATA promoters.

TABLE S13
SENSITIVITY TO CORE-PROMOTER ELEMENTS, RELATIVE TO DISTAL SEQUENCE

Architecture	Element	Null	Baseline	GCR
Transformer	TATA	1.14 ± 0.09	2.13 ± 1.43	23.98 ± 6.99
	BREu	2.63 ± 0.30	2.94 ± 0.86	2.42 ± 0.75
	BREd	2.02 ± 0.19	2.46 ± 0.25	2.08 ± 0.28
	Inr	6.22 ± 0.82	5.81 ± 0.28	5.82 ± 0.63
CNN-GlobalPool	TATA	2.17 ± 0.37	3.56 ± 1.26	18.70 ± 2.46
	BREu	3.02 ± 0.37	2.76 ± 0.38	2.68 ± 0.31
	BREd	1.97 ± 0.19	2.10 ± 0.42	2.62 ± 0.28
	Inr	1.11 ± 0.07	1.03 ± 0.07	1.08 ± 0.09
CNN-Dense	TATA	2.58 ± 0.17	3.72 ± 0.18	12.74 ± 0.99
	BREu	2.73 ± 0.05	2.90 ± 0.10	2.05 ± 0.17
	BREd	2.82 ± 0.15	3.07 ± 0.16	1.96 ± 0.12
	Inr	2.28 ± 0.14	2.45 ± 0.17	1.76 ± 0.01
NT-v2-50M	TATA	1.81 ± 0.25	1.33 ± 0.03	19.01 ± 4.44
	BREu	2.00 ± 0.07	1.67 ± 0.17	2.65 ± 0.28
	BREd	1.60 ± 0.06	1.54 ± 0.14	2.10 ± 0.05
	Inr	1.58 ± 0.06	1.44 ± 0.08	1.50 ± 0.05
NT-v2-100M	TATA	1.45 ± 0.36	3.73 ± 0.65	19.10 ± 3.15
	BREu	1.54 ± 0.12	1.95 ± 0.23	2.87 ± 0.30
	BREd	1.36 ± 0.14	1.67 ± 0.11	2.44 ± 0.34
	Inr	1.12 ± 0.06	1.39 ± 0.16	1.54 ± 0.07
NT-v2-500M-LoRA	TATA	1.67 ± 0.20	2.70 ± 0.61	19.20 ± 1.18
	BREu	1.36 ± 0.04	1.50 ± 0.05	2.73 ± 0.14
	BREd	1.08 ± 0.08	1.25 ± 0.06	2.54 ± 0.19
	Inr	1.04 ± 0.07	1.13 ± 0.04	2.23 ± 0.21
DeepSTARR	TATA	3.04 ± 0.58	7.15 ± 0.89	5.12 ± 0.47
	BREu	2.79 ± 0.18	3.31 ± 0.31	3.49 ± 0.17
	BREd	2.58 ± 0.45	3.79 ± 0.61	2.38 ± 0.30
	Inr	2.17 ± 0.32	2.53 ± 0.17	1.91 ± 0.21

Mean absolute change in logit over all single-nucleotide substitutions in the element, divided by the same quantity at +61 to +100 in the same promoters (76 probe promoters, seeds 0–2). TATA: the 7-bp TBP core; BREu and BREd: the 7 bp immediately upstream and downstream of it; Inr: -2 to $+4$. A value of 1 means no more sensitive than distal sequence. Only TATA is protected by GCR.